```
---
AWSTemplateFormatVersion: '2010-09-09'
Description: 'AWS in Action: chapter 11 (minimal)'
Parameters:                                          ①
  VPC:
    Type: 'AWS::EC2::VPC::Id'
  SubnetA:
    Type: 'AWS::EC2::Subnet::Id'
  SubnetB:
    Type: 'AWS::EC2::Subnet::Id'
Resources:
  CacheSecurityGroup:                                ②
    Type: 'AWS::EC2::SecurityGroup'
    Properties:
      GroupDescription: cache
      VpcId: !Ref VPC
      SecurityGroupIngress:
      - IpProtocol: tcp                              ③
        FromPort: 6379
        ToPort: 6379
        CidrIp: '0.0.0.0/0'
  CacheSubnetGroup:                                  ④
    Type: 'AWS::ElastiCache::SubnetGroup'
    Properties:
      Description: cache
      SubnetIds:
      - Ref: SubnetA                                 ⑤
      - Ref: SubnetB
  Cache:
    Type: 'AWS::ElastiCache::CacheCluster'           ⑥
    Properties:
      CacheNodeType: 'cache.t2.micro'                ⑦
      CacheSubnetGroupName: !Ref CacheSubnetGroup
      Engine: redis                                  ⑧
      NumCacheNodes: 1                               ⑨
      VpcSecurityGroupIds:
      - !Ref CacheSecurityGroup
```

① Defines VPC and subnets as parameters

② The security group to manage which traffic is allowed to enter/leave the cluster

③ Redis listens on port 6379. This allows access from all IP addresses, but because the cluster has only private IP addresses, access is possible only from inside the VPC. You will improve this in section 11.3.

④ Subnets are defined within a subnet group (same approach is used in RDS).

⑤ List of subnets that can be used by the cluster

⑥ The resource to define the Redis cluster

⑦ The node type cache.t2.micro comes with 0.555 GiB memory and is part of the Free Tier.

⑧ ElastiCache supports redis and memcached. We are using redis because we want to make use of advanced data structures supported only by Redis.

⑨ Creates a single-node cluster for testing, which is not recommended for production workloads because it introduces a single point of failure